

CALC I CHEAT SHEET — PAGE 1 (EXAMPLES ONLY)

Limits — Worked Examples

1. **Rational limit at infinity** $\lim_{x \rightarrow \infty} \frac{5x^3 - 2x}{2x^3 + 7} = \frac{5}{2}$.
2. **Rationalizing trick** $\lim_{x \rightarrow 0} \frac{\sqrt{4+x} - 2}{x} = \frac{1}{\sqrt{4+x} + 2} \rightarrow \frac{1}{4}$.
3. **Squeeze theorem** $\lim_{x \rightarrow 0} x^2 \sin(1/x) = 0$.
4. **Oscillation (no limit)** $\lim_{x \rightarrow 0} \sin(1/x)$ does not exist.
5. **Piecewise continuity** $f(x) = \begin{cases} x^2 & x < 1 \\ 3x - 2 & x \geq 1 \end{cases}$
 $\lim_{x \rightarrow 1} f(x) = 1$.
6. **Exponential vs polynomial** $\lim_{x \rightarrow \infty} \frac{x^5}{e^x} = 0$.
7. **Log vs power** $\lim_{x \rightarrow \infty} \frac{\ln x}{x^{1/3}} = 0$.
8. **Trig limit** $\lim_{x \rightarrow 0} \frac{\sin(7x)}{x} = 7$.
9. **Root limit** $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 3x} - x) = \frac{3}{2}$.
10. **Hard algebraic limit** $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} = 12$.

L'Hôpital — Worked Examples

1. **Classic 0/0** $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.
2. **Infinity/infinity** $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$.
3. **0 · ∞ form** $x \ln x = \frac{\ln x}{1/x} \rightarrow \text{L'Hôpital} \rightarrow 0$.
4. **1[∞] form** $(1 + 2x)^{1/x} \rightarrow \ln y = \frac{\ln(1+2x)}{x} \rightarrow e^2$.
5. **∞ - ∞ form** $\sqrt{x^2 + 3x} - x \rightarrow \text{multiply by conjugate} \rightarrow \frac{3}{2}$.
6. **Log trick** $\lim_{x \rightarrow \infty} (1 + \frac{3}{x})^x = e^3$.

Derivative Examples (No Rules Shown)

1. **Product + chain** $f(x) = x^2 e^{3x}$ $f'(x) = e^{3x}(2x + 3x^2)$.
2. **Quotient example** $f(x) = \frac{3x^2 - 1}{x + 2}$ $f'(x) = \frac{6x(x+2) - (3x^2 - 1)}{(x+2)^2}$.
3. **Trig composite** $f(x) = \sin(x^3)$ $f'(x) = 3x^2 \cos(x^3)$.
4. **Mixed composite** $f(x) = \ln(\cos(2x))$ $f'(x) = -2 \tan(2x)$.
5. **Inverse trig** $f(x) = \arctan(5x)$ $f'(x) = \frac{5}{1 + 25x^2}$.
6. **Exponential composite** $f(x) = e^{\sin x}$ $f'(x) = e^{\sin x} \cos x$.
7. **Log composite** $f(x) = \ln(x^2 + 4)$ $f'(x) = \frac{2x}{x^2 + 4}$.
8. **Hard composite** $f(x) = \sqrt{1 + \tan(3x)}$ $f'(x) = \frac{3 \sec^2(3x)}{2\sqrt{1 + \tan(3x)}}$.

CALC I CHEAT SHEET — PAGE 2 (EXAMPLES ONLY)

FTC Examples

1. **FTC1 — derivative of an integral** $F(x) = \int_2^x \sin(t^2) dt$
 $F'(x) = \sin((x^3)^2) \cdot 3x^2 = 3x^2 \sin(x^6)$.
2. **FTC2 — evaluate definite integral** $\int_0^3 (2x + 1) dx = [x^2 + x]_0^3 = 12$.
3. **Variable bounds (top function)** $F(x) = \int_1^{x^2} \ln t dt$
 $F'(x) = \ln(x^2) \cdot 2x$.
4. **Variable bounds (bottom function)** $G(x) = \int_x^4 e^{t^2} dt$
 $G'(x) = -e^{x^2}$.
5. **Both bounds variable** $H(x) = \int_x^{x^3} \cos t dt$ $H'(x) = \cos(x^3) \cdot 3x^2 - \cos(x)$.
6. **Accumulation function** $A(x) = \int_0^x \sqrt{1 + t^2} dt$ $A'(x) = \sqrt{1 + x^2}$.

Implicit Differentiation

1. **Circle** $x^2 + y^2 = 25 \rightarrow y' = -x/y$.
2. **Hard implicit** $x^3 + y^3 = 6xy$ Differentiate: $3x^2 + 3y^2 y' = 6(y + xy')$ Solve: $y' = \frac{6y - 3x^2}{3y^2 - 6x}$.
3. **Trig implicit** $\sin(xy) = x \cos(xy)(y + xy') = 1$ $y' = \frac{1 - y \cos(xy)}{x \cos(xy)}$.

Logarithmic Differentiation

1. **Power to power** $y = x^x \rightarrow \ln y = x \ln x \rightarrow y' = x^x (\ln x + 1)$.
2. **Ugly product** $y = (x^2 + 1)^5 (\sin x)^3$ $\ln y = 5 \ln(x^2 + 1) + 3 \ln(\sin x)$ Differentiate and multiply by y .

Linear Approximation

1. **Approximate** $\sqrt{10} f(x) = \sqrt{x}$, $a = 9$, $f'(x) = \frac{1}{2\sqrt{x}}$.
 $L(x) = 3 + \frac{1}{6}(x - 9) \rightarrow L(10) = 3 + \frac{1}{6} = 3.1667$.
2. **Approximate** $e^{0.1} f(x) = e^x$, $a = 0$. $L(x) = 1 + x \rightarrow L(0.1) = 1.1$.

Mean Value Theorem

1. **Simple MVT** $f(x) = x^2 + 3x$ on $[0, 4]$. $f'(x) = 2x + 3$.
Solve $f'(c) = \frac{f(4) - f(0)}{4 - 0} = \frac{28}{4} = 7$. $2c + 3 = 7 \rightarrow c = 2$.
2. **Trig MVT** $f(x) = \sin x$ on $[0, \pi]$. Slope = $\frac{0 - 0}{\pi} = 0$. Solve $\cos c = 0 \rightarrow c = \pi/2$.

Derivative Traps

1. $|x|$ nondifferentiable Corner at $x = 0$.
2. **Cusp example** $f(x) = x^{2/3} \rightarrow f'(x) = \frac{2}{3}x^{-1/3}$ undefined at 0.

U-Substitution Examples

1. **Polynomial inside power** $\int (3x^2)(x^3 + 1)^5 dx$ $u = x^3 + 1 \rightarrow du = 3x^2 dx = \frac{u^6}{6} + C$.
2. **Trig composite** $\int \cos(5x) dx$ $u = 5x \rightarrow du = 5dx = \frac{1}{5} \sin(5x) + C$.
3. **Exponential composite** $\int x e^{x^2} dx$ $u = x^2 \rightarrow du = 2x dx = \frac{1}{2} e^{x^2} + C$.
4. **Rational substitution** $\int \frac{1}{x \ln x} dx$ $u = \ln x \rightarrow du = \frac{1}{x} dx = \ln |\ln x| + C$.
5. **Completing square** $\rightarrow \arctan \int \frac{1}{x^2 + 4x + 13} dx$ $x^2 + 4x + 13 = (x + 2)^2 + 9 = \frac{1}{3} \arctan \frac{x+2}{3} + C$.
6. **Hard exponential** $\int e^{3x+1} dx = \frac{1}{3} e^{3x+1} + C$.
7. **Reverse chain rule trap** $\int x^2 \sqrt{x^3 + 5} dx$ $u = x^3 + 5 \rightarrow du = 3x^2 dx = \frac{2}{9} (u)^{3/2} + C$.

8. **Weird substitution** $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$.

Inverse Trig Integrals

1. **Arctan form** $\int \frac{1}{x^2+9} dx = \frac{1}{3} \arctan(x/3) + C$.
2. **Arcsin form** $\int \frac{1}{\sqrt{25-x^2}} dx = \arcsin(x/5) + C$.
3. **Arcsec form** $\int \frac{1}{|x|\sqrt{x^2-4}} dx = (|x|/2) + C$.
4. **Completing square** $\rightarrow$ **arctan** $\int \frac{1}{(x-1)^2+16} dx = \frac{1}{4} \arctan \frac{x-1}{4} + C$.

Definite Integral Examples

1. **Simple polynomial** $\int_0^2 (3x^2 - x) dx = [x^3 - \frac{1}{2}x^2]_0^2 = 6$.
2. **Trig definite** $\int_0^{\pi/2} \sin x dx = 1$.
3. **Exponential definite** $\int_0^1 e^{2x} dx = \frac{1}{2}(e^2 - 1)$.
4. **Absolute value** $\int_{-3}^2 |x| dx = \int_{-3}^0 (-x) dx + \int_0^2 x dx = \frac{9}{2} + 2$.

Area Between Curves (with diagrams)

Template: Area = $\int_a^b [\text{top} - \text{bottom}] dx$.

1. **Vertical slicing** Between $y = x^2$ and $y = 4$:

$$A = \int_{-2}^2 (4 - x^2) dx = \frac{32}{3}.$$

CALC I CHEAT SHEET — PAGE 3 (EXAMPLES ONLY)

Optimization Templates + Examples

General Template:

- Identify quantity to optimize (A, V, cost, distance).
- Write all constraints.
- Express target in one variable.
- Differentiate, set derivative = 0.
- Check endpoints if needed.

1. **Max area rectangle under parabola** Maximize area of rectangle under $y = 9 - x^2$. Width = $2x$, height = $9 - x^2$. $A(x) = 2x(9 - x^2) = 18x - 2x^3$. $A'(x) = 18 - 6x^2 = 0 \rightarrow x = \sqrt{3}$. Max area = $12\sqrt{3}$.

2. **Min distance to a point** Minimize distance from $(0,0)$ to curve $y = 4 - x^2$. $D^2 = x^2 + (4 - x^2)^2$. Differentiate: $2x + 2(4 - x^2)(-2x) = 0$. Solve for $x \rightarrow$ gives minimum.

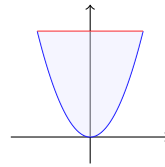
3. **Open-top box from sheet** Square cut size x from 12×20 sheet. $V = x(12 - 2x)(20 - 2x)$. Differentiate, solve $V' = 0$.

4. **Cylinder with fixed surface area** $S = 2\pi rh + 2\pi r^2 =$ constant. Solve h in terms of r , plug into $V = \pi r^2 h$. Differentiate $\rightarrow h = 2r$.

5. **Profit maximization** Revenue $R(x) = 50x - 0.1x^2$, cost $C(x) = 10x + 200$. Profit $P(x) = R - C = 40x - 0.1x^2 - 200$. $P'(x) = 40 - 0.2x = 0 \rightarrow x = 200$.

Related Rates (with diagrams)

General Template:



2. **Horizontal slicing** Between $x = y^2$ and $x = 4$:

$$A = \int_0^2 (4 - y^2) dy.$$

3. **Switching order** Region between $y = x$ and $y = \sqrt{x}$: Intersection at $x = 0, 1$.

$$A = \int_0^1 (\sqrt{x} - x) dx.$$

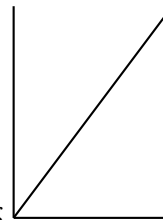
4. **Piecewise area** Between $y = |x|$ and $y = 2$: Break at $x = 0$.

$$A = \int_{-2}^0 (2 - (-x)) dx + \int_0^2 (2 - x) dx.$$

Integration Traps

1. **Forgetting +C** $\int \frac{1}{x} dx = \ln |x| + C$.
2. **Wrong substitution direction** If $u = g(x)$, then $du = g'(x)dx$ — must match exactly.
3. **Splitting logs incorrectly** $\ln(a + b) \neq \ln a + \ln b$.

- Draw diagram.
- Label variables.
- Write equation relating variables.
- Differentiate w.r.t. time.
- Plug in values at the instant.



1. **Ladder sliding** 10 ft ladder: $x^2 + y^2 = 100$. Differentiate: $2xx' + 2yy' = 0$. Solve for y' .

2. **Cone filling** $V = \frac{1}{3}\pi r^2 h$, and r/h constant. Differentiate: $V' = \frac{1}{3}\pi(2rr'h + r^2h')$. Use ratio to eliminate r' .

3. **Shadow problem** Similar triangles: $\frac{y}{x} = \frac{6}{x+d}$. Differentiate both sides.

4. **Sphere inflating** $V = \frac{4}{3}\pi r^3$. $V' = 4\pi r^2 r'$.

Curve Sketching (Examples + Diagrams)

General Template:

- Find domain.

- Find $f' \rightarrow$ critical points $\rightarrow$ sign chart.
- Find $f'' \rightarrow$ concavity $\rightarrow$ inflection points.
- Asymptotes if rational.

1. Sign chart example $f(x) = x^3 - 3x$. $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$. Critical points: $x = \pm 1$.

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Increasing on $(-1, 1)$, decreasing elsewhere.

2. Concavity example $f''(x) = 6x$. Concave down for $x < 0$, up for $x > 0$. Inflection at $x = 0$.

3. Rational asymptotes $f(x) = \frac{2x^2+1}{x-1}$. Vertical asymptote: $x = 1$. Long division $\rightarrow$ slant asymptote $y = 2x + 2$.

MVT Example

$f(x) = x^3 - 4x$ on $[-2, 2]$. Slope $= \frac{f(2)-f(-2)}{4} = \frac{0-0}{4} = 0$. Solve $f'(c) = 0$: $f'(x) = 3x^2 - 4 = 0 \rightarrow x = \pm \frac{2}{\sqrt{3}}$.

CALC I CHEAT SHEET — PAGE 4 (GENERAL STRATEGIES)

General Limit Strategies

1. If you see $0/0$:

- Factor
- Rationalize
- L'Hôpital
- Expand small-angle trig: $\sin x \approx x$, $\cos x \approx 1 - x^2/2$

2. If you see ∞/∞ :

- Divide by highest power
- L'Hôpital
- Compare growth: $\ln x \ll x^a \ll a^x$

3. If you see $\infty - \infty$:

- Combine into one fraction
- Multiply by conjugate

4. If you see $0 \cdot \infty$: Rewrite as $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

5. If you see 1^∞ : Take $\ln$ and convert to $\frac{0}{0}$.

Derivative Recognition Patterns

1. Chain rule triggers:

- Something inside something: $\sin(3x)$, $\ln(x^2 + 1)$
- Root of a function: $\sqrt{g(x)}$

2. Product rule triggers:

- $x \cdot \text{trig}$
- $x \cdot e^x$
- polynomial $\cdot \log$

3. Quotient rule triggers:

Linear Approx Example

Approximate $\sqrt[3]{8.5}$. $f(x) = x^{1/3}$, $a = 8$. $f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{12}$. $L(x) = 2 + \frac{1}{12}(x-8)$. $L(8.5) = 2 + \frac{1}{12}(0.5) = 2.0417$.

Application Traps

1. Wrong variable elimination Always eliminate using the constraint before differentiating.

2. Related rates sign errors If a length is decreasing, its derivative is negative.

3. Curve sketching trap Critical points where $f' = 0$ are not always maxima/minima — check sign chart.

- Rational functions where simplifying first is messy

4. Implicit triggers:

- xy , x/y , $\sin(xy)$, $x^2 + y^2$

5. Log-diff triggers:

- Products of many factors
- Powers to powers: x^x , $(\sin x)^x$

Integral Recognition Patterns

1. U-sub triggers:

- Inside function + its derivative: $f'(x)f(x)^n$
- Exponential with linear inside: e^{ax+b}
- Trig composite: $\cos(5x)$, $\sin(3x)$

2. Inverse trig triggers:

- $\frac{1}{a^2+x^2} \rightarrow \arctan$
- $\frac{1}{\sqrt{a^2-x^2}} \rightarrow \arcsin$

- Completing the square $\rightarrow \arctan$

3. Absolute value integrals: Break at zeros of inside.

Optimization Master Template

Steps:

- Draw picture
- Identify quantity to optimize
- Write constraint
- Eliminate variables
- Differentiate
- Solve $f' = 0$
- Check endpoints

Common shapes:

- Rectangle: $A = lw$
- Box: $V = lwh$
- Cylinder: $V = \pi r^2 h$
- Distance: $D^2 = (x - a)^2 + (y - b)^2$

Related Rates Master Template

Steps:

- Draw diagram
- Label variables
- Write relationship equation
- Differentiate w.r.t. time
- Plug in values at instant

Common relationships:

- Pythagorean: $x^2 + y^2 = c^2$
- Similar triangles
- Volume formulas
- Area formulas

Curve Sketching Checklist

1. Domain **2. Intercepts** **3. Asymptotes** **4. First derivative:** critical points, increasing/decreasing **5. Second derivative:** concavity, inflection points **6. End behavior** **7. Sketch**

Trig Quick Reference (NOT identities)

Signs by quadrant:

- Q1: all +
- Q2: sin +, cos -, tan -
- Q3: sin -, cos -, tan +
- Q4: sin -, cos +, tan -

Unit circle memory:

- $\frac{\sqrt{3}}{2} \approx 0.866$
- $\frac{\sqrt{2}}{2} \approx 0.707$
- $\frac{1}{2} = 0.5$

Algebra Tricks That Save Time

1. Rationalizing: multiply by conjugate. **2. Factoring:** pull out highest power. **3. Completing square:** for arctan integrals. **4. Avoid quotient rule:** simplify first. **5. Avoid product rule:** factor constants, rewrite.

What To Try When Stuck

- Rewrite the function (logs, exponents, trig)
- Factor something out
- Substitute u for something messy
- Check if derivative/integral resembles a known pattern
- Draw a picture
- Check endpoints
- Check signs of derivatives
- Try L'Hôpital if limit is indeterminate

Common Exam Traps

1. Forgetting domain restrictions **2. Forgetting $+C$** **3. Misidentifying max/min without sign chart** **4. Wrong variable eliminated in optimization** **5. Related rates sign errors** **6. Using quotient rule when simplification is easier** **7. Mixing up arctan and arcsin forms** **8. Forgetting to check endpoints**